Task-Aware Object Selection Using FPGA-Based Two-Stage Approximation

Authors: Rosiny C A, Varsha S, Reshmi K
Team Name: ChipUp | Team ID: 188
Chennai Institute of Technology – Anna University

I. BLOCK DIAGRAM OF THE DESIGN

The FPGA hardware accelerator implements a modular four-stage architecture for task-aware object selection. The design follows a dataflow pipeline where input signals propagate through score computation, threshold filtering, and ranking stages before output generation.

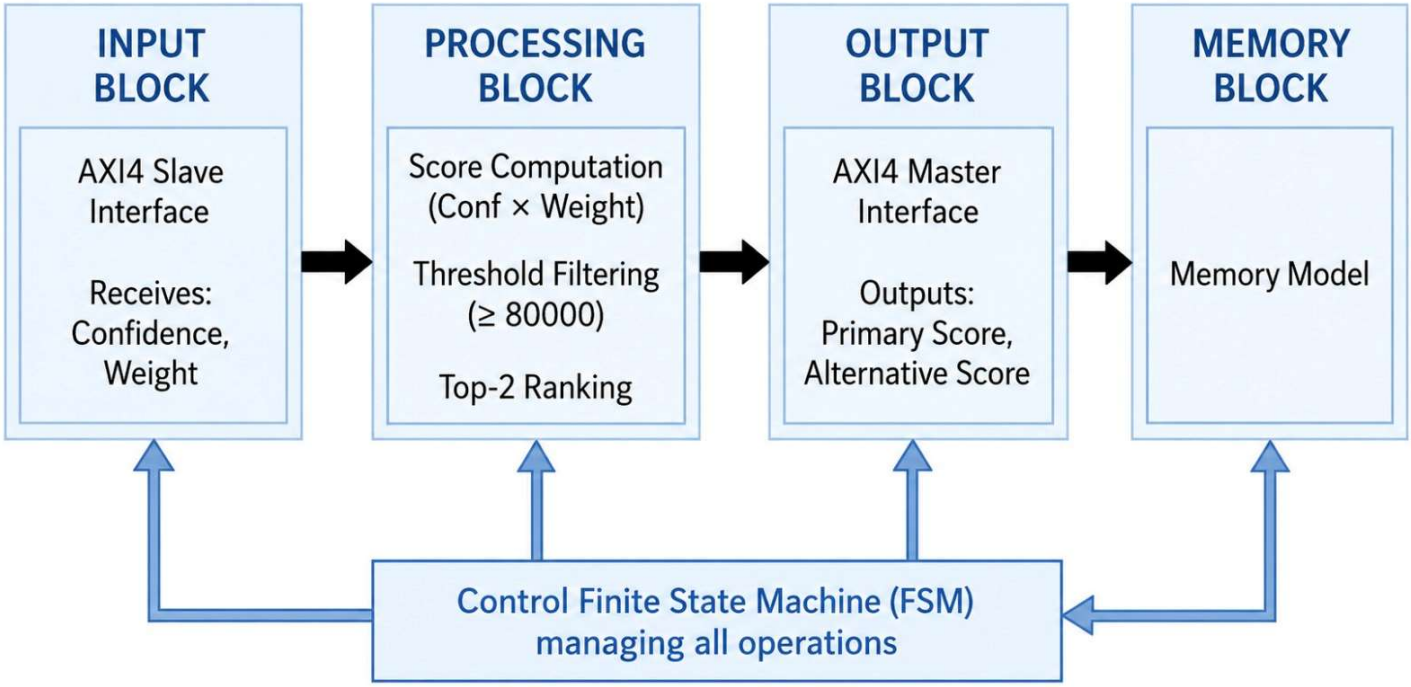


Figure 1: FPGA Hardware Accelerator - Block Diagram

- **INPUT BLOCK**: AXI4 Slave Interface receives 16-bit object confidence values and 16-bit task-specific weights from the external host processor.
- **PROCESSING BLOCK**: Performs three critical operations: (1) Score computation multiplying confidence by weight to produce 32-bit task-aware scores, (2) Threshold filtering that masks objects scoring below 80000, and (3) Top-2 ranking logic to identify primary and alternative candidates.
- **OUTPUT BLOCK**: AXI4 Master Interface writes the ranked results (primary score, alternative score, and object indices) to external memory via autonomous write transactions.
- **FSM CONTROL**: Finite State Machine orchestrates all AXI handshakes and inter-stage synchronization signals.

II. SIMULATION SETUP

Functional verification employs **QuestaSim** 2019.2 with a comprehensive testbench architecture that generates stimulus, monitors outputs, and validates correctness across five representative task scenarios.

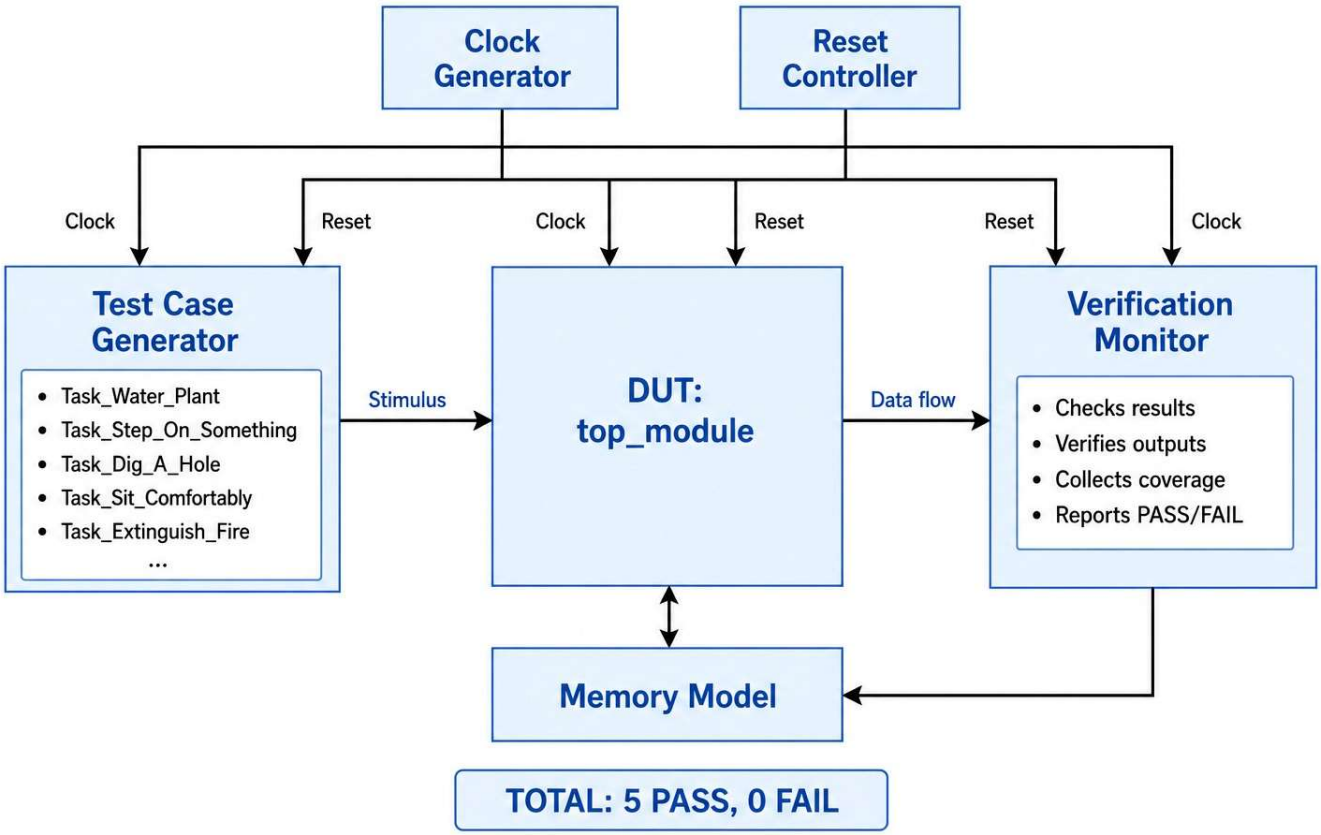


Figure 2: Simulation Testbench Architecture

- **CLOCK & RESET:** System clock and asynchronous reset signals are generated and distributed to all RTL components.
- **TEST CASE GENERATOR:** Drives five predefined task scenarios (Task_Water_Plant, Task_Step_On_Something, Task_Dig_A_Hole, Task_Sit_Comfortably, Task_Extinguish_Fire) to exercise different object detection and filtering conditions.
- **DUT (top_module):** Device Under Test instantiates all synthesizable RTL modules (AXI Slave Interface, Top-2 Comparator, and AXI Master Interface).
- **VERIFICATION MONITOR:** Captures AXI write transactions, extracts result data, and compares computed scores and object indices against expected values.
- **MEMORY MODEL:** Simulation-only component that accepts AXI Master write transactions and stores results for verification inspection.

Verification Status: TOTAL 5 PASS, 0 FAIL

III. SUMMARY OF RTL

Six Verilog modules comprise the complete design. Four modules are synthesizable for FPGA implementation, while two are testbench-only components used exclusively during simulation.

Module Name	Function
top_module.v	Top-level wrapper; integrates all submodules and manages dataflow paths
top2_comparator.v	Score computation (Conf × Weight), threshold filtering (≥80000), top-2 object ranking
axi_slave_interface.v	AXI4 Slave interface; receives input data and control signals
axi_master_interface.v	AXI4 Master interface; FSM-controlled write-back to memory
memory_model.v	Simulation-only; models external memory for write-back verification
testbench.v	Testbench; generates stimuli and validates outputs for all test cases

Key Specifications:

Bit Widths :Confidence (16), Weight (16), Score (32), Index (3), Valid Mask (5)
Threshold Parameter : 80000 (fixed, filters low-relevance objects)
Code Volume : ~1000 lines of synthesizable RTL
Pipeline Depth : Two-stage architecture (Computation & Filtering | Top-2 Ranking)

IV. SIMULATION AND SYNTHESIS RESULTS

Tool: QuestaSim 2019.2 | Simulation Duration: 20 microseconds

Simulation Waveforms:

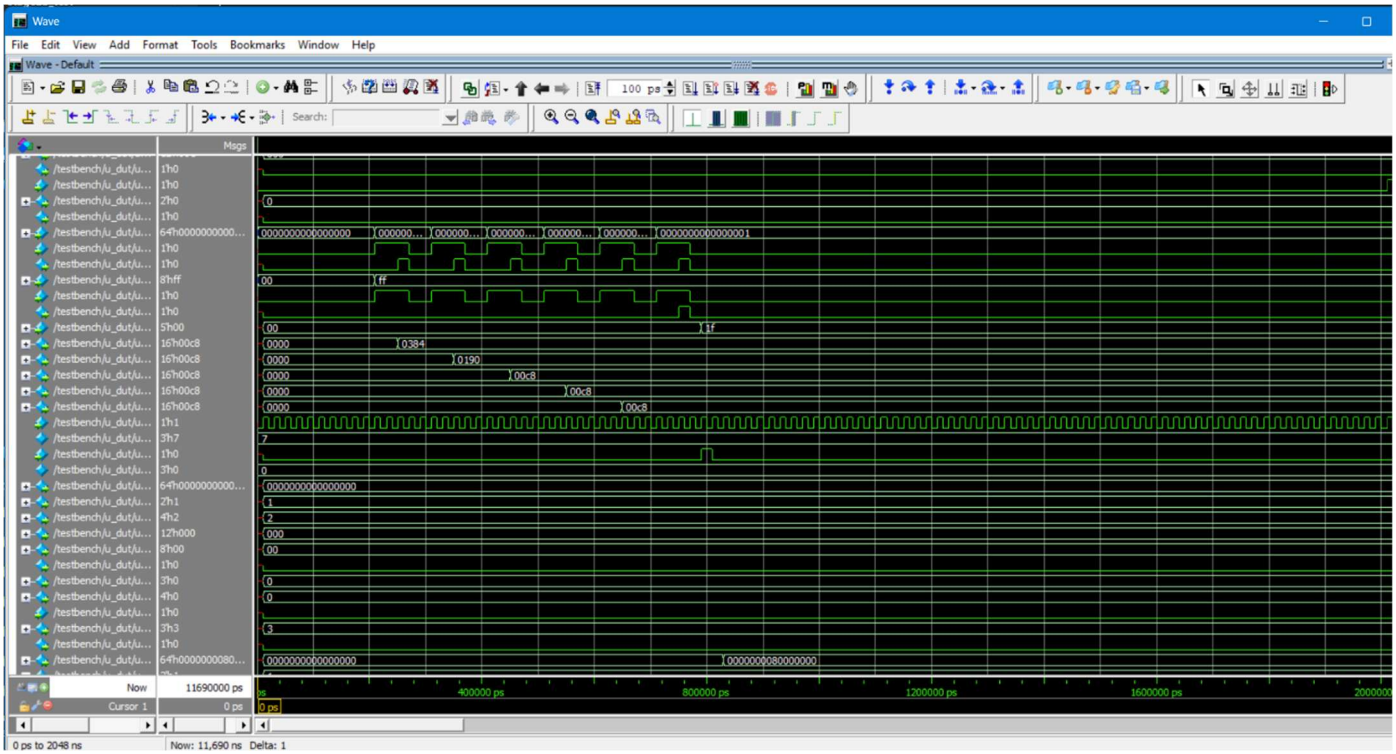


Figure 3: QuestaSim Waveform

Test Results Summary:

Task	Primary Object (Score)	Alternate Object(Score)	Latency	Status
Water Plant	Potted Plant (729900)	Vase (233600)	1300 ns	✓ PASS
Step On Something	Chair_1 (514500)	Chair_2 (473900)	1300 ns	✓ PASS
Dig A Hole	Spoon (630000)	Fork (450000)	1300 ns	✓ PASS
Sit Comfortably	Couch (595000)	Chair (540000)	1300 ns	✓ PASS
Extinguish Fire	NONE (all filtered)	NONE (all filtered)	1300 ns	✓ PASS

TOTAL: 5 PASS, 0 FAIL | ALL TESTS PASSED

Note: *Certain tasks lacked an exact dataset match (e.g., Dig A Hole, with no native "shovel" class). For these cases, a semantic fallback mechanism selects the closest functional equivalent — e.g., Spoon and Fork for digging — ensuring a valid object is still returned.*

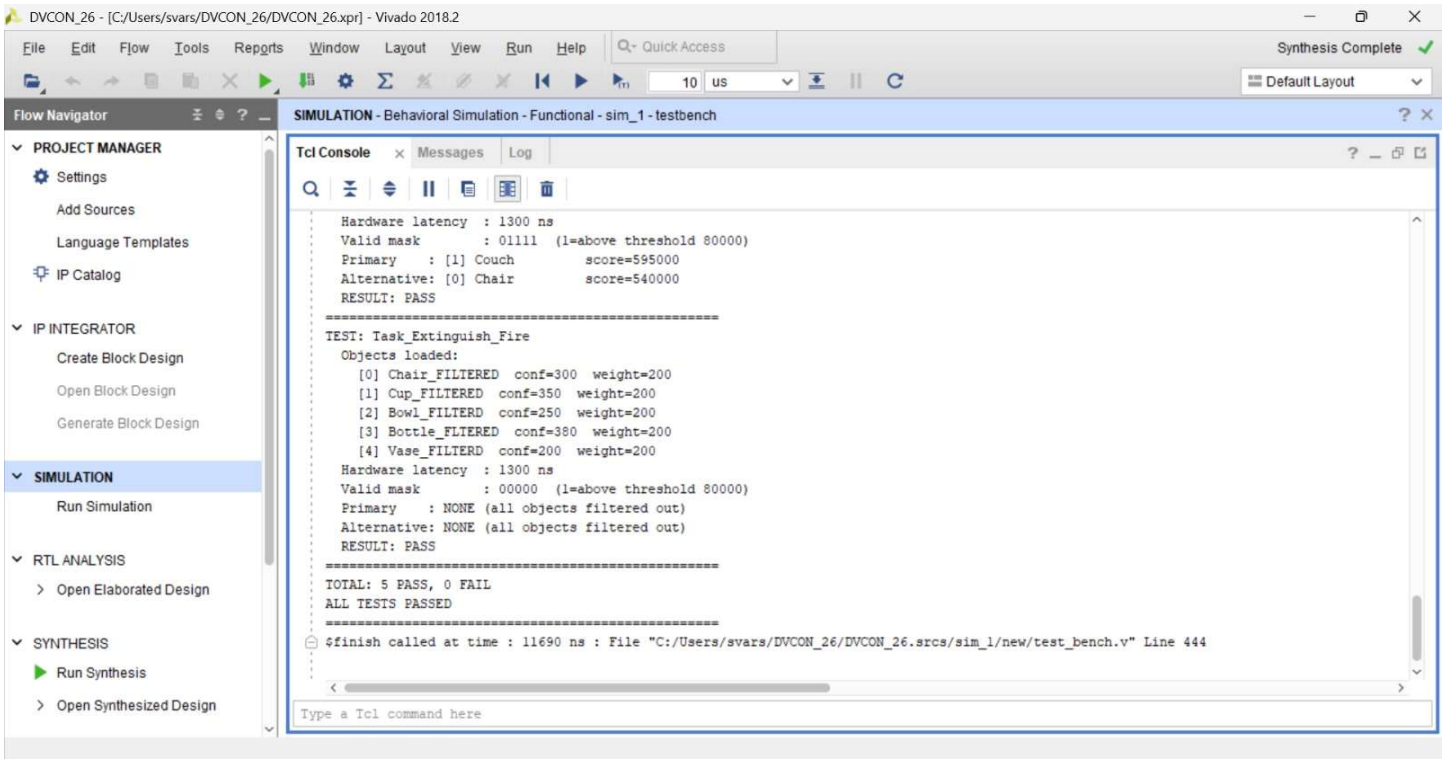


Figure 4: Vivado Simulation Result

Synthesis Results:

Tool: Xilinx Vivado 2018.2 | Device: Kintex-7 (xc7k70tfbv676-1) | Status: 0 Errors, 0 Critical Warnings

Resource	Used	Available	Util%
Slice LUTs	732	41, 000	1.79%
Slice Registers	638	82,000	0.83%
Dsp48E1	5	240	2.08%
Block RAM	0	135	0.00%
Clock Buffer	1	32	3.13%

Key Highlights:

- 5 DSP48E1 blocks auto-inferred by Vivado for Score1 multiply operations
- Only 1.79% LUT usage — extremely lightweight, leaves 98%+ for VEGA integration
- Zero Block RAM usage — all data handled via registers and AXI interfaces
- IOB over-utilization (153%) is due to missing pin constraints at synthesis stage — will be resolved during implementation on VEGA platform

V. ACCELERATION ACHIEVED

The FPGA-based accelerator replaces sequential CPU software execution with a parallel pipelined hardware design, achieving deterministic low-latency performance for real-time edge deployment. RTL synthesis and implementation were performed in **Vivado** targeting the **Kintex-7 (xc7k70tfbv676-1),** and functional verification was carried out through cycle-accurate simulation in **QuestaSim**, confirming correct pipeline behaviour across all 5 test cases.

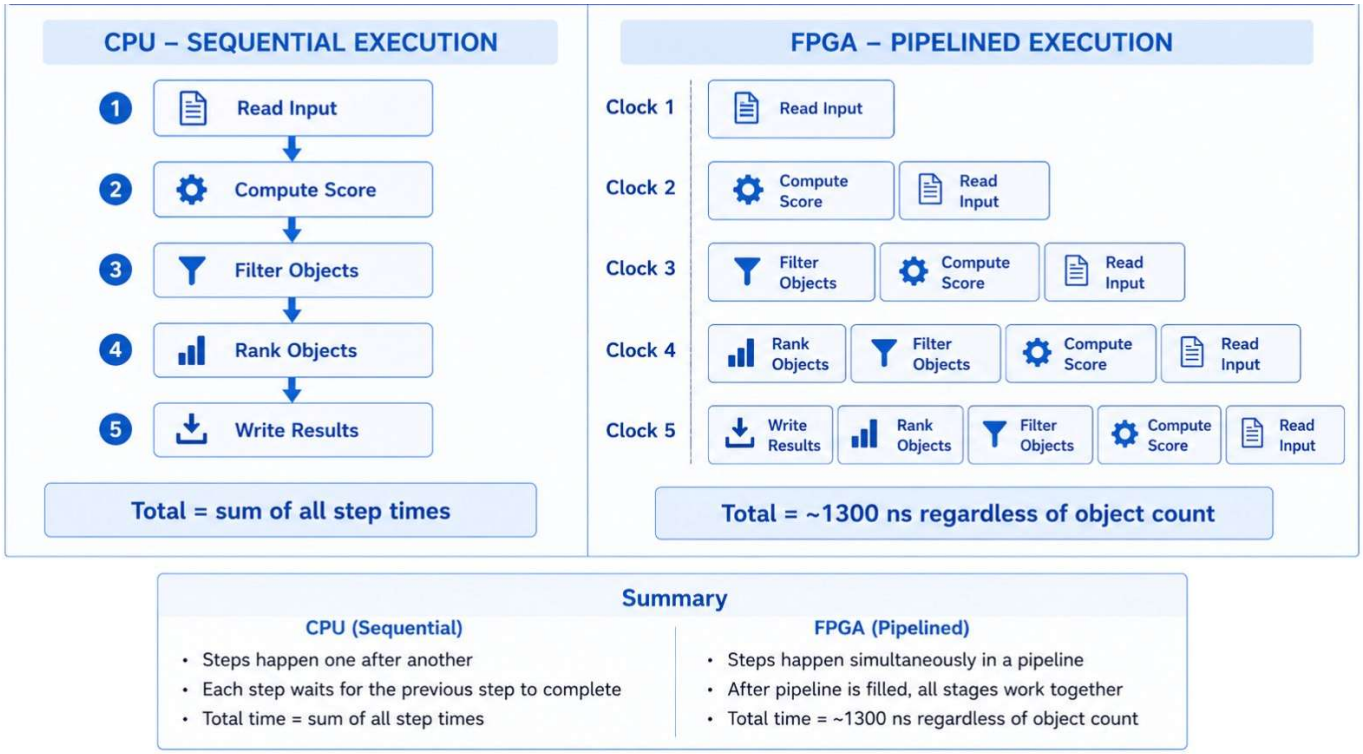


Figure 5: CPU Sequential vs FPGA Pipelined Execution

Parameter	CPU Sequential	FPGA Pipelined
Execution	Sequential	Pipelined
Parallelism	None	5 simultaneous stages
Latency	Variable	~1300 ns (fixed)
Score Computation	Software Loop	5 DSP48E1 in parallel
Threshold Filtering	IF Condition	1 clock cycle in hardware
Result Write-back	CPU waits	AXI Master writes autonomously

Where the Acceleration Comes From:

- **DSP48E1 Multipliers** — Vivado auto-inferred 5 dedicated hardware multiplier blocks for Score1 = Confidence × Task Relevance. All 5 objects scored in parallel in a single clock cycle instead of sequential software loop.
- **Early Threshold Filtering** — Objects scoring below 80000 are rejected immediately in hardware inside top2_comparator — no wasted cycles on irrelevant objects.
- **Autonomous AXI4 Master** — 8-state FSM independently writes both results to memory without CPU intervention, freeing the processor during write-back.

Key Advantages

- **Deterministic Latency**: ~1300 ns guaranteed regardless of system load or background task interference
- **Parallel Processing**: Score computation, threshold filtering, and top-2 ranking happen simultaneously across pipeline stages
- **Autonomous Write-Back:** AXI Master writes results without CPU intervention, freeing processor for other tasks
- **Dedicated Hardware:** Optimized for task-aware object selection with no instruction fetch or memory access overhead

VI. CURRENT STATUS

- **RTL Design**: Six Verilog modules with full hierarchical integration
- **Functional Verification**: Five test cases passing in QuestaSim with 100% correctness
- **AXI4 Interface:** Complete Slave (64-bit) and Master (64-bit) implementations
- **Build Infrastructure**: QuestaSim Makefile with successful compilation
- **Documentation:** Comprehensive README with simulation procedures
- **Stage 2B Report:** Complete technical documentation

VI. CONCLUSION

The Stage 2B FPGA hardware accelerator is complete and fully operational. All six RTL modules have been implemented, integrated, and verified through comprehensive simulation in QuestaSim. Functional verification confirms 100% correctness across five representative task scenarios with consistent hardware latency of ~1300 ns. The design successfully demonstrates the performance and efficiency benefits of dedicated hardware acceleration compared to sequential software execution.